AmirPouya Moeini

Ph.D. Student @ University of Cambridge, Trinity Hall College

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Education

Ph.D. in Engineering — Oct 2023 – Present

University of Cambridge, UK — Probabilistic Systems, Information, and Inference Group (PSI^2)
Supervisor: Prof. Albert Guillén i Fàbregas
Advisor: Prof. Ramji Venkataramanan

B.Sc. in Computer Engineering (Minor in Mathematics) — Sep 2018 – Jun 2023

Sharif University of Technology, Iran
Thesis: *Information-theoretic bounds for online learning* (20/20)

Research Interests

Applied Probability {high-dimensional probability, concentration properties}, **Information Theory** {Shannon theory, information-theoretic methods in inference}, and **Combinatorics** {probabilistic combinatorics, graph theory}

Publications

(J1). "The Expurgated Error Exponent is Not Universally Achievable" [Link]
Moeini, A., *Dalai, M., and Guillén i Fàbregas, A., Submitted to IEEE Transactions on Information Theory*

(C4). "Random Gilbert-Varshamov Codes for Joint Source-Channel Coding" [Link]
Moeini, A. *and Guillén i Fàbregas, A., IEEE International Symposium on Information Theory* ***(ISIT 2026)***

(C3). "Two-Class Joint Source-Channel Coding: Expurgated Exponents with i.i.d. Distributions" [Link]
Moeini, A. *and Guillén i Fàbregas, A., International Zurich Seminar on Information and Communication* ***(IZS 2026)***

(C2). "Dual-Domain Exponent of Maximum Mutual Information Decoding" [Link]
Moeini, A. *and Guillén i Fàbregas, A., IEEE Information Theory Workshop* ***(ITW 2025)***

(C1). "Class-Based Expurgation Attains Csiszár's Expurgated Source-Channel Exponent" [Link]
Moeini, A. *and Guillén i Fàbregas, A., IEEE Information Theory Workshop* ***(ITW 2025)***

Academic Visits

National University of Singapore, Singapore — October 2025
Institute of Data Science – School of Computing, visiting Dr. Jonathan Scarlett

University of Brescia, Brescia, Italy — September 2025
Department of Information Engineering, visiting Dr. Marco Dalai

Honours and Awards

2023	**Cambridge International Trust – Full Scholarship** *(≈£62,000 per year)*	Cambridge, UK
2022	**Summer AScI @ Aalto University Fellowship**	Espoo, Finland
2018	**Gold Medal** *International Olympiad on Astronomy and Astrophysics (IOAA)*	Beijing, China
2017	**Gold Medal** *National Olympiad on Astronomy and Astrophysics*	Iran

Teaching Experience

Supervision University of Cambridge

- Information Theory and Coding (3F7), Michaelmas 24, 25
- Data Transmission (3F4), Lent 25, 26

Teaching Assistant Sharif University of Technology

- Information theory & Coding*, Fall 22
- Theoretical Machine learning*, Fall 22
- Statistical Machine learning*, Spring 22
- Mathematical Statistics, Fall 21
- Data Structure & Algorithm, Spring 21
- Linear algebra, Spring 21
- Regression Analysis, Spring 21
- Artificial intelligence, Fall 21, Spring 22

*: Graduate-level (Ph.D.) Course

Selected Courses

University of Cambridge

Computational statistics and machine learning, Advanced information theory and coding

Sharif University of Technology

Probability

Information theory and coding*, Information-theoretic methods in high-dimensional statistics*, High-dimensional probability*, Graphical models*, Stochastic processes, Bayesian methods in statistics*, Index coding

Mathematics

Real analysis, Algebra, Linear algebra, Applied linear algebra, Probability & statistics

Machine Learning

Machine learning theory*, Artificial intelligence, Computer simulation, Modern information retrieval

Computer Science

Design algorithms, Data structures and algorithms, Compiler design, Database design

Audited Courses

Probabilistic combinatorics, Entropy methods in combinatorics, Ramsey theory, Random discrete structures, Stochastic calculus and applications to finance (*Cambridge*)

Research Experience

Probabilistic Machine Learning Group, Aalto University, Finland Jul 2022 – Oct 2022

- Approximate Bayesian Computation in the Presence of Outliers
- Supervisor: Prof. Samuel Kaski

- Developing computationally efficient algorithms that enhance the robustness of Approximate Bayesian Computation (ABC) inference under outliers and model misspecification.

Machine Learning Laboratory, Sharif University of Technology, Iran Oct 2021 – Mar 2023

- Information-Theoretic Bounds for Online Learning Problems
- Supervisors: Dr. Mahdieh Soleymani and Dr. Hassan Hafez-Kolahi

- We introduce information-theoretic tools for online learning with adversarial data and derive performance upper bounds via conditional mutual information.

Work Experience

Data Scientist Intern | *Comma AI-Med* Mar 2023 - Jun 2023

Breast Cancer Diagnosis Enhancement via State-of-the-Art Generative Models in Mammography

Data Engineer Intern | *Institute for Research in Fundamental Sciences (IPM)* Jun 2020 - Sep 2020

Contributed to the R&D team of Iran's National Observatory (INO) project

- Weather station data pipe-line and data analysis
- Night-sky cloud prediction

Skills

Programming Language: Python (*advanced*), Java (*advanced*), C / C++, SQL.

Statistical Programming Languages: R (*advanced*), Julia (*advanced*), Stata, MatLab & Octave.

Tools and Software: PyTorch, LaTeX, Jupyter NB / G Colab, Git / GitHub

Language: Farsi/Persian (*native*), English (*fluent*), French (*basic*).